LECTURE 04

Sets

Definitions, operations, solved examples and exercises

Overview

This chapter deals with the concept of a set and operations on sets. The concept of sets will be useful in studying relations and functions.

Sets and their representations

> A **set** is a well-defined collection of objects.

There are two methods of representing a set:
(i) Roster or tabular form (ii) Set-builder form

The empty set

> A set which does not contain any element is called the **empty set**, the **void set** or the **null set**, and is denoted by $\{\}$ or $\varnothing$.

Finite and infinite sets

A set which consists of a finite number of elements is called a **finite set**; otherwise, the set is called an **infinite set**.

Subsets

> A set A is said to be a **subset** of a set B if every element of A is also an element of B. In symbols, we write $A \subset B$ if $a \in A \Rightarrow a \in B$.

We denote

the set of real numbers by	$\mathbb{R}$
the set of natural numbers by	$\mathbb{N}$
the set of integers by	$\mathbb{Z}$
the set of rational numbers by	$\mathbb{Q}$
the set of irrational numbers by	$\mathbb{T}$

We observe that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}, \quad \mathbb{T} \subset \mathbb{R}, \quad \mathbb{Q} \not\subset \mathbb{T}, \quad \mathbb{N} \not\subset \mathbb{T}.$$

Equal sets

Given two sets A and B, if every element of A is also an element of B and every element of B is also an element of A, then the sets A and B are said to be **equal**. Two equal sets have exactly the same elements.

Intervals as subsets of $\mathbb{R}$

Let $a, b \in \mathbb{R}$ and $a < b$. Then

(a) An **open interval**, denoted by (a, b), is the set of real numbers $\{x : a < x < b\}$.
(b) A **closed interval**, denoted by $[a, b]$, is the set of real numbers $\{x : a \leq x \leq b\}$.
(c) Intervals closed at one end and open at the other are given by

$$[a, b) = \{x : a \leq x < b\}, \quad (a, b] = \{x : a < x \leq b\}.$$

Power set

The collection of all subsets of a set A is called the **power set** of A, denoted by $P(A)$. If the number of elements in A is n, i.e. $n(A) = n$, then the number of elements in $P(A)$ is 2^n.

Universal set

This is a basic set in a particular context, whose elements and subsets are relevant to that context. For example, for the set of vowels of the English alphabet, the universal set can be the set of all letters of the English alphabet. The universal set is denoted by U.

Venn diagrams

Venn diagrams are diagrams which represent the relationship between sets. For example, the set of natural numbers is a subset of the set of whole numbers, which is a subset of the integers. We can represent this relationship by the Venn diagram in Fig 1.1.

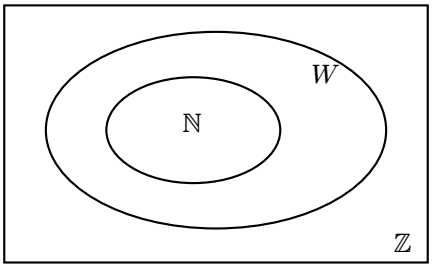


Fig 1.1 $\mathbb{N} \subset W \subset \mathbb{Z}$

Operations on Sets

Union of sets

The **union** of two sets A and B is the set C which consists of all those elements which are either in A or in B. In symbols,

$$C = A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

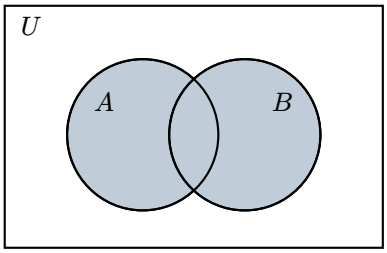


Fig 1.2 (a) $A \cup B$

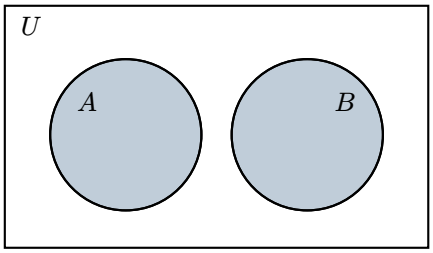


Fig 1.2 (b) $A \cup B$

Some properties of the operation of union:
(i) $A \cup B = B \cup A$
(ii) $(A \cup B) \cup C = A \cup (B \cup C)$
(iii) $A \cup \varnothing = A$
(iv) $A \cup A = A$
(v) $U \cup A = U$

Intersection of sets

The **intersection** of two sets A and B is the set which consists of all those elements which belong to both A and B. Symbolically,

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

When $A \cap B = \varnothing$, A and B are called **disjoint sets**.

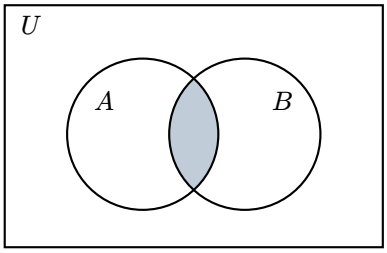


Fig 1.3 (a) $A \cap B$

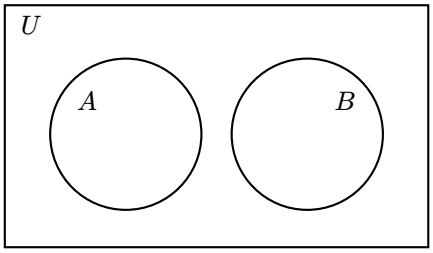


Fig 1.3 (b) $A \cap B = \varnothing$

Some properties of the operation of intersection:

(i) $A \cap B = B \cap A$
(ii) $(A \cap B) \cap C = A \cap (B \cap C)$
(iii) $\varnothing \cap A = \varnothing$; $U \cap A = A$
(iv) $A \cap A = A$
(v) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
(vi) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Difference of sets

The **difference** of two sets A and B, denoted by $A - B$, is the set of elements which belong to A but not to B. We write

$$A - B = \{x : x \in A \text{ and } x \notin B\}, \quad B - A = \{x : x \in B \text{ and } x \notin A\}.$$

Complement of a set

Let U be the universal set and A a subset of U. The **complement** of A is the set of all elements of U which are not elements of A. Symbolically,

$$A' = \{x : x \in U \text{ and } x \notin A\}, \quad \text{also} \quad A' = U - A.$$

Some properties of the complement of sets:

(i) **Law of complements:** (a) $A \cup A' = U$ (b) $A \cap A' = \varnothing$
(ii) **De Morgan's laws:** (a) $(A \cup B)' = A' \cap B'$ (b) $(A \cap B)' = A' \cup B'$
(iii) $(A')' = A$
(iv) $U' = \varnothing$ and $\varnothing' = U$

Counting formulae for union and intersection

Let A, B and C be any finite sets. Then

(a) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
(b) If $A \cap B = \varnothing$, then $n(A \cup B) = n(A) + n(B)$
(c) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Solved Examples

Short answer type

Example 1

Write the following sets in roster form.

(i) $A = \{x \mid x \text{ is a positive integer less than 10 and } 2^x - 1 \text{ is an odd number}\}$

(ii) $C = \{x : x^2 + 7x - 8 = 0, x \in \mathbb{R}\}$

Solution.

(i) $2^x - 1$ is always an odd number for all positive integral values of x. In particular, $2^x - 1$ is odd for $x = 1, 2, ..., 9$. Thus $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

(ii) $x^2 + 7x - 8 = 0$, i.e. $(x + 8)(x - 1) = 0$, giving $x = -8$ or $x = 1$. Thus $C = \{-8, 1\}$.

Example 2

State which of the following statements are true and which are false. Justify your answer.

(i) $37 \notin \{x \mid x \text{ has exactly two positive factors}\}$

(ii) $28 \in \{y \mid \text{the sum of all the positive factors of } y \text{ is } 2y\}$

(iii) $7747 \in \{t \mid t \text{ is a multiple of } 37\}$

Solution.

(i) **False.** Since 37 has exactly two positive factors, 1 and 37, 37 belongs to the set.

(ii) **True.** The sum of the positive factors of 28 is $1 + 2 + 4 + 7 + 14 + 28 = 56 = 2(28)$.

(iii) **False.** 7747 is not a multiple of 37.

Example 3

If X and Y are subsets of the universal set U, show that

(i) $Y \subset X \cup Y$ (ii) $X \cap Y \subset X$ (iii) $X \subset Y \Rightarrow X \cap Y = X$

Solution.

(i) $X \cup Y = \{x \mid x \in X \text{ or } x \in Y\}$. Thus $x \in Y \Rightarrow x \in X \cup Y$. Hence $Y \subset X \cup Y$.

(ii) $X \cap Y = \{x \mid x \in X \text{ and } x \in Y\}$. Thus $x \in X \cap Y \Rightarrow x \in X$. Hence $X \cap Y \subset X$.

(iii) Note that $x \in X \cap Y \Rightarrow x \in X$, so $X \cap Y \subset X$. Also, since $X \subset Y$,

$$x \in X \Rightarrow x \in Y \Rightarrow x \in X \cap Y,$$

so that $X \subset X \cap Y$. Hence $X = X \cap Y$.

Example 4

Given that $\mathbb{N} = \{1, 2, 3, ..., 100\}$,

(i) write the subset A of $\mathbb{N}$ whose elements are odd numbers;

(ii) write the subset B of $\mathbb{N}$ whose elements are represented by $x + 2$, where $x \in \mathbb{N}$.

Solution.

(i) $A = \{x \mid x \in \mathbb{N} \text{ and } x \text{ is odd}\} = \{1, 3, 5, 7, ..., 99\}$.

(ii) $B = \{y \mid y = x + 2, x \in \mathbb{N}\}$. For $1 \in \mathbb{N}$, $y = 1 + 2 = 3$; for $2 \in \mathbb{N}$, $y = 2 + 2 = 4$; and so on. Therefore $B = \{3, 4, 5, 6, ..., 100\}$.

Example 5

Given that $E = \{2, 4, 6, 8, 10\}$. If n represents any member of E, write the following sets containing all numbers represented by

(i) $n + 1$ (ii) n^2

Solution.

(i) Let $A = \{x \mid x = n + 1, n \in E\}$. For $2 \in E$, $x = 3$; for $4 \in E$, $x = 5$; and so on. Therefore $A = \{3, 5, 7, 9, 11\}$.

(ii) Let $B = \{x \mid x = n^2, n \in E\}$. For $2 \in E$, $x = 2^2 = 4$; for $4 \in E$, $x = 4^2 = 16$; for $6 \in E$, $x = 6^2 = 36$; and so on. Hence $B = \{4, 16, 36, 64, 100\}$.

Example 6

Let $X = \{1, 2, 3, 4, 5, 6\}$. If n represents any member of X, express the following as sets:

(i) $n \in X$ but $2n \notin X$ (ii) $n + 5 = 8$ (iii) n is greater than 4

Solution.

(i) Let $A = \{x \mid x \in X \text{ and } 2x \notin X\}$. Now

$$\begin{array}{lll} 1 \notin A & \text{as} & 2 \cdot 1 = 2 \in X \\ 2 \notin A & \text{as} & 2 \cdot 2 = 4 \in X \\ 3 \notin A & \text{as} & 2 \cdot 3 = 6 \in X \\ 4 \in A & \text{as} & 2 \cdot 4 = 8 \notin X \\ 5 \in A & \text{as} & 2 \cdot 5 = 10 \notin X \\ 6 \in A & \text{as} & 2 \cdot 6 = 12 \notin X \end{array}$$

So $A = \{4, 5, 6\}$.

(ii) Let $B = \{x \mid x \in X \text{ and } x + 5 = 8\}$. Here $B = \{3\}$, as $x = 3 \in X$ and $3 + 5 = 8$, and there is no other element of X such that $x + 5 = 8$.

(iii) Let $C = \{x \mid x \in X, x > 4\}$. Therefore $C = \{5, 6\}$.

Example 7

Draw Venn diagrams to illustrate the following relationships among the sets E, M and U, where E is the set of students studying English in a school, M is the set of students studying Mathematics in the same school, and U is the set of all students in that school.

(i) All the students who study Mathematics study English, but some students who study English do not study Mathematics.

(ii) There is no student who studies both Mathematics and English.

(iii) Some of the students study Mathematics but not English, some study English but not Mathematics, and some study both.

(iv) Not all students study Mathematics, but every student studying English studies Mathematics.

Solution.

(i) Since all of the students who study Mathematics study English, but some students who study English do not study Mathematics, $M \subset E \subset U$ (Fig 1.4).

(ii) Since there is no student who studies both English and Mathematics, $E \cap M = \varnothing$ (Fig 1.5).

(iii) Since some students study both English and Mathematics, some English only and some Mathematics only, the Venn diagram is Fig 1.6.

(iv) Since every student studying English studies Mathematics, $E \subset M \subset U$ (Fig 1.7).

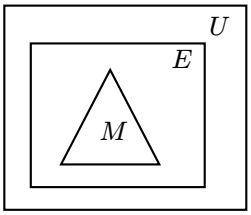


Fig 1.4

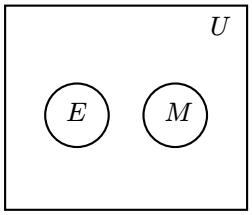


Fig 1.5

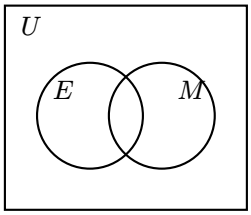


Fig 1.6

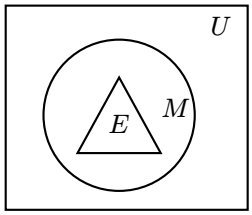


Fig 1.7

Example 8

For all sets A, B and C, is $(A \cap B) \cup C = A \cap (B \cup C)$? Justify your statement.

Solution. No. Consider the sets $A = \{1, 2, 3\}$, $B = \{2, 3, 5\}$, $C = \{4, 5, 6\}$. Then

$$(A \cap B) \cup C = (\{1, 2, 3\} \cap \{2, 3, 5\}) \cup \{4, 5, 6\} = \{2, 3\} \cup \{4, 5, 6\} = \{2, 3, 4, 5, 6\},$$
$$A \cap (B \cup C) = \{1, 2, 3\} \cap (\{2, 3, 5\} \cup \{4, 5, 6\}) = \{1, 2, 3\} \cap \{2, 3, 4, 5, 6\} = \{2, 3\}.$$

Therefore $(A \cap B) \cup C \neq A \cap (B \cup C)$.

Example 9

Use the properties of sets to prove that for all sets A and B,

$$A - (A \cap B) = A - B.$$

Solution. We have

$$\begin{aligned} A - (A \cap B) &= A \cap (A \cap B)' && \text{(since } A - B = A \cap B') \\ &= A \cap (A' \cup B') && \text{(by De Morgan's law)} \\ &= (A \cap A') \cup (A \cap B') && \text{(by the distributive law)} \\ &= \varnothing \cup (A \cap B') \\ &= A \cap B' = A - B. \end{aligned}$$

Long answer type

Example 10

For all sets A, B and C, is $(A - B) \cap (C - B) = (A \cap C) - B$? Justify your answer.

Solution. **Yes.** Let $x \in (A - B) \cap (C - B)$. Then

$$\begin{aligned} & x \in A - B \text{ and } x \in C - B \\ \Rightarrow\ & (x \in A \text{ and } x \notin B) \text{ and } (x \in C \text{ and } x \notin B) \\ \Rightarrow\ & (x \in A \text{ and } x \in C) \text{ and } x \notin B \\ \Rightarrow\ & x \in A \cap C \text{ and } x \notin B \\ \Rightarrow\ & x \in (A \cap C) - B. \end{aligned}$$

So $(A - B) \cap (C - B) \subset (A \cap C) - B$. (1)

Conversely, let $y \in (A \cap C) - B$. Then

$$\begin{aligned} & y \in A \cap C \text{ and } y \notin B \\ \Rightarrow\ & (y \in A \text{ and } y \in C) \text{ and } y \notin B \\ \Rightarrow\ & (y \in A \text{ and } y \notin B) \text{ and } (y \in C \text{ and } y \notin B) \\ \Rightarrow\ & y \in A - B \text{ and } y \in C - B \\ \Rightarrow\ & y \in (A - B) \cap (C - B). \end{aligned}$$

So $(A \cap C) - B \subset (A - B) \cap (C - B)$. (2)

From (1) and (2), $(A - B) \cap (C - B) = (A \cap C) - B$.

Example 11

Let A, B and C be sets. Show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Solution. We first show that $A \cup (B \cap C) \subset (A \cup B) \cap (A \cup C)$. Let $x \in A \cup (B \cap C)$. Then

$$\begin{aligned} & x \in A \text{ or } x \in B \cap C \\ \Rightarrow\ & x \in A \text{ or } (x \in B \text{ and } x \in C) \\ \Rightarrow\ & (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\ \Rightarrow\ & x \in A \cup B \text{ and } x \in A \cup C \\ \Rightarrow\ & x \in (A \cup B) \cap (A \cup C). \end{aligned}$$

Thus $A \cup (B \cap C) \subset (A \cup B) \cap (A \cup C)$. (1)

Now we show that $(A \cup B) \cap (A \cup C) \subset A \cup (B \cap C)$. Let $x \in (A \cup B) \cap (A \cup C)$. Then

$$\begin{aligned} & x \in A \cup B \text{ and } x \in A \cup C \\ \Rightarrow\ & (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\ \Rightarrow\ & x \in A \text{ or } (x \in B \text{ and } x \in C) \\ \Rightarrow\ & x \in A \text{ or } x \in B \cap C \\ \Rightarrow\ & x \in A \cup (B \cap C). \end{aligned}$$

Thus $(A \cup B) \cap (A \cup C) \subset A \cup (B \cap C)$. (2)

So, from (1) and (2), $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Example 12

Let P be the set of prime numbers and let $S = \{t \mid 2^t - 1 \text{ is a prime}\}$. Prove that $S \subset P$.

Solution. The contrapositive of $x \in S \Rightarrow x \in P$ is $x \notin P \Rightarrow x \notin S$. We prove this contrapositive by contradiction.

Let $x \notin P$, so x is a composite number. Assume that $x \in S$. Then $2^x - 1 = m$, where m is a prime number, i.e. $2^x = m + 1$. This is not true for composite numbers; for example, for $x = 4$, $2^4 = 16$, which cannot be equal to the sum of a prime number m and 1. Thus we arrive at a contradiction, and $x \notin S$.

Hence, whenever $x \notin P$, we have $x \notin S$. So $S \subset P$.

Example 13

Of 50 students taking examinations in Mathematics, Physics and Chemistry, each student passed in at least one of the subjects: 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry, and at most 20 Physics and Chemistry. What is the largest possible number that could have passed all three examinations?

Solution. Let M, P and C be the sets of students passing in Mathematics, Physics and Chemistry respectively. Then

$$n(M \cup P \cup C) = 50, \quad n(M) = 37, \quad n(P) = 24, \quad n(C) = 43,$$
$$n(M \cap P) \leq 19, \quad n(M \cap C) \leq 29, \quad n(P \cap C) \leq 20.$$

Now

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C) \leq 50$$

$$\Rightarrow 37 + 24 + 43 - 19 - 29 - 20 + n(M \cap P \cap C) \leq 50$$
$$\Rightarrow n(M \cap P \cap C) \leq 50 - 36$$
$$\Rightarrow n(M \cap P \cap C) \leq 14.$$

Thus the largest possible number that could have passed all three examinations is 14.

Objective type questions

Choose the correct answer from the four given options in each of Examples 14 to 16.

Example 14

Each set X_r contains 5 elements and each set Y_r contains 2 elements, and $\bigcup_{r=1}^{20} X_r = S = \bigcup_{r=1}^{n} Y_r$. If each element of S belongs to exactly 10 of the X_r's and to exactly 4 of the Y_r's, then n is

(A) 10 (B) 20 (C) 100 (D) 50

Solution. The correct answer is **(B)**. Since $n(X_r) = 5$ and $\bigcup_{r=1}^{20} X_r = S$, counting with repetition gives $20 \times 5 = 100$ elements. But each element of S belongs to exactly 10 of the X_r's, so $100/10 = 10$ is the number of distinct elements in S.

Also, each element of S belongs to exactly 4 of the Y_r's and each Y_r contains 2 elements. If there are n sets Y_r, then

$$\frac{2n}{4} = 10, \quad \text{which gives} \quad n = 20.$$

Example 15

Two finite sets have m and n elements respectively. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. The values of m and n respectively are

(A) $7, 6$ (B) $5, 1$ (C) $6, 3$ (D) $8, 7$

Solution. The correct answer is **(C)**. Let A and B be such sets, i.e. $n(A) = m$, $n(B) = n$. Then $n(P(A)) = 2^m$ and $n(P(B)) = 2^n$, so

$$2^m - 2^n = 56 \Rightarrow 2^n(2^{m-n} - 1) = 2^3 \times 7$$
$$\Rightarrow n = 3, \quad 2^{m-n} - 1 = 7$$
$$\Rightarrow m = 6.$$

Example 16

The set $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$ is equal to

(A) $B \cap C'$ (B) $A \cap C$ (C) $B \cup C'$ (D) $A \cap C'$

Solution. The correct choice is **(A)**, since

$$\begin{aligned}(A\cup B\cup C)\cap(A\cap B'\cap C')'\cap C' &= (A\cup(B\cup C))\cap(A'\cup(B\cup C))\cap C'\\ &= ((A\cap A')\cup(B\cup C))\cap C'\\ &= (\varnothing\cup(B\cup C))\cap C'\\ &= (B\cap C')\cup(C\cap C') = (B\cap C')\cup\varnothing = B\cap C'.\end{aligned}$$

Fill in the blanks

Example 17

If A and B are two finite sets, then $n(A)+n(B)$ is equal to _____.

Solution. Since $n(A\cup B)=n(A)+n(B)-n(A\cap B)$,

$$n(A)+n(B)=n(A\cup B)+n(A\cap B).$$

Example 18

If A is a finite set containing n elements, then the number of subsets of A is _____.

Solution. 2^n.

True or false

Example 19

Let R and S be the sets defined as follows:

$$R=\{x\in\mathbb{Z}\mid x \text{ is divisible by } 2\},\quad S=\{y\in\mathbb{Z}\mid y \text{ is divisible by } 3\}.$$

Then $R\cap S=\varnothing$.

Solution. **False**, since 6 is divisible by both 2 and 3. Thus $R\cap S\neq\varnothing$.

Example 20

$\mathbb{Q}\cap\mathbb{R}=\mathbb{Q}$, where $\mathbb{Q}$ is the set of rational numbers and $\mathbb{R}$ is the set of real numbers.

Solution. **True**, since $\mathbb{Q}\subset\mathbb{R}$, so $\mathbb{Q}\cap\mathbb{R}=\mathbb{Q}$.

Exercises

Short answer type

1. Write the following sets in roster form.
   (i) $A=\{x : x\in\mathbb{R}, 2x+11=15\}$ (ii) $B=\{x\mid x^2=x, x\in\mathbb{R}\}$
   (iii) $C=\{x\mid x$ is a positive factor of a prime number $p\}$
2. Write the following sets in roster form.
   (i) $D=\{t\mid t^3=t, t\in\mathbb{R}\}$ (ii) $E=\left\{w\mid \frac{w-2}{w+3}=3, w\in\mathbb{R}\right\}$
   (iii) $F=\{x\mid x^4-5x^2+6=0, x\in\mathbb{R}\}$
3. If $Y=\{x\mid x$ is a positive factor of the number $2^{p-1}(2^p-1)$, where 2^p-1 is a prime number $\}$, write Y in roster form.
4. State which of the following statements are true and which are false. Justify your answer.

(i) $35 \in \{x \mid x \text{ has exactly four positive factors}\}$
(ii) $128 \in \{y \mid \text{the sum of all the positive factors of } y \text{ is } 2y\}$
(iii) $3 \notin \{x \mid x^4 - 5x^3 + 2x^2 - 112x + 6 = 0\}$
(iv) $496 \notin \{y \mid \text{the sum of all the positive factors of } y \text{ is } 2y\}$

5. Given $L = \{1, 2, 3, 4\}$, $M = \{3, 4, 5, 6\}$ and $N = \{1, 3, 5\}$, verify that $L - (M \cup N) = (L - M) \cap (L - N)$.

6. If A and B are subsets of the universal set U, show that
(i) $A \subset A \cup B$ (ii) $A \subset B \Leftrightarrow A \cup B = B$ (iii) $(A \cap B) \subset A$

7. Given that $\mathbb{N} = \{1, 2, 3, ..., 100\}$, write
(i) the subset of $\mathbb{N}$ whose elements are even numbers;
(ii) the subset of $\mathbb{N}$ whose elements are perfect squares.

8. If $X = \{1, 2, 3\}$ and n represents any member of X, write the following sets containing all numbers represented by
(i) $4n$ (ii) $n + 6$ (iii) $\frac{n}{2}$ (iv) $n - 1$

9. If $Y = \{1, 2, 3, ..., 10\}$ and a represents any element of Y, write the following sets, containing all the elements satisfying the given conditions.
(i) $a \in Y$ but $a^2 \notin Y$ (ii) $a + 1 = 6$, $a \in Y$ (iii) a is less than 6 and $a \in Y$

10. A, B and C are subsets of the universal set U. If $A = \{2, 4, 6, 8, 12, 20\}$, $B = \{3, 6, 9, 12, 15\}$, $C = \{5, 10, 15, 20\}$ and U is the set of all whole numbers, draw a Venn diagram showing the relation of U, A, B and C.

11. Let U be the set of all boys and girls in a school, G the set of all girls in the school, B the set of all boys in the school, and S the set of all students in the school who take swimming. Some, but not all, students in the school take swimming. Draw a Venn diagram showing one of the possible interrelationships among the sets U, G, B and S.

12. For all sets A, B and C, show that $(A - B) \cap (A - C) = A - (B \cup C)$.

Determine whether each of the statements in Exercises 13–17 is true or false. Justify your answer.

13. For all sets A and B, $(A - B) \cup (A \cap B) = A$.

14. For all sets A, B and C, $A - (B - C) = (A - B) - C$.

15. For all sets A, B and C, if $A \subset B$, then $A \cap C \subset B \cap C$.

16. For all sets A, B and C, if $A \subset B$, then $A \cup C \subset B \cup C$.

17. For all sets A, B and C, if $A \subset C$ and $B \subset C$, then $A \cup B \subset C$.

Using properties of sets, prove the statements given in Exercises 18–21.

18. For all sets A and B, $A \cup (B - A) = A \cup B$.

19. For all sets A and B, $A - (A - B) = A \cap B$.

20. For all sets A and B, $A - (A \cap B) = A - B$.

21. For all sets A and B, $(A \cup B) - B = A - B$.

22. Let $T = \left\{x \mid \frac{x+5}{x-7} - 5 = \frac{4x-40}{13-x}\right\}$. Is T an empty set? Justify your answer.

Long answer type

23. Let A, B and C be sets. Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

24. Out of 100 students, 15 passed in English, 12 passed in Mathematics, 8 in Science, 6 in English and Mathematics, 7 in Mathematics and Science, 4 in English and Science, and 4 in all three. Find how many passed
    (i) in English and Mathematics but not in Science
    (ii) in Mathematics and Science but not in English
    (iii) in Mathematics only
    (iv) in more than one subject only

25. In a class of 60 students, 25 students play cricket, 20 students play tennis, and 10 students play both games. Find the number of students who play neither.

26. In a survey of 200 students of a school, it was found that 120 study Mathematics, 90 study Physics, 70 study Chemistry, 40 study Mathematics and Physics, 30 study Physics and Chemistry, 50 study Chemistry and Mathematics, and 20 study none of these subjects. Find the number of students who study all three subjects.

27. In a town of 10 000 families it was found that 40% of families buy newspaper A, 20% buy newspaper B, 10% buy newspaper C, 5% buy A and B, 3% buy B and C, and 4% buy A and C. If 2% of families buy all three newspapers, find
    (a) the number of families which buy newspaper A only;
    (b) the number of families which buy none of A, B and C.

28. In a group of 50 students, the numbers of students studying French, English and Sanskrit were found to be as follows: French $= 17$, English $= 13$, Sanskrit $= 15$, French and English $= 9$, English and Sanskrit $= 4$, French and Sanskrit $= 5$, English, French and Sanskrit $= 3$. Find the number of students who study
    (i) French only
    (ii) English only
    (iii) Sanskrit only
    (iv) English and Sanskrit but not French
    (v) French and Sanskrit but not English
    (vi) French and English but not Sanskrit
    (vii) at least one of the three languages
    (viii) none of the three languages

Objective type questions

Choose the correct answer from the four given options in each of Exercises 29 to 43.

29. Suppose $A_1, A_2, ..., A_{30}$ are thirty sets each having 5 elements and $B_1, B_2, ..., B_n$ are n sets each with 3 elements. Let $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S$, and each element of S belongs to exactly 10 of the A_i's and exactly 9 of the B_j's. Then n is equal to
    (A) 15 (B) 3 (C) 45 (D) 35

30. Two finite sets have m and n elements. The number of subsets of the first set is 112 more than that of the second set. The values of m and n are, respectively,
    (A) $4, 7$ (B) $7, 4$ (C) $4, 4$ (D) $7, 7$

31. The set $(A \cap B')' \cup (B \cap C)$ is equal to
    (A) $A' \cup B \cup C$ (B) $A' \cup B$ (C) $A' \cup C'$ (D) $A' \cap B$

32. Let F_1 be the set of parallelograms, F_2 the set of rectangles, F_3 the set of rhombuses, F_4 the set of squares and F_5 the set of trapeziums in a plane. Then F_1 may be equal to
    (A) $F_2 \cap F_3$
    (B) $F_3 \cap F_4$
    (C) $F_2 \cup F_5$
    (D) $F_2 \cup F_3 \cup F_4 \cup F_1$

33. Let S be the set of points inside a square, T the set of points inside a triangle and C the set of points inside a circle. If the triangle and the circle intersect each other and are contained in the square, then
    (A) $S \cap T \cap C = \varnothing$
    (B) $S \cup T \cup C = C$
    (C) $S \cup T \cup C = S$
    (D) $S \cup T = S \cap C$

34. Let R be the set of points inside a rectangle with sides a and b $(a, b > 1)$, with two sides along the positive directions of the x-axis and y-axis. Then
(A) $R = \{(x, y) : 0 \le x \le a, 0 \le y \le b\}$ (B) $R = \{(x, y) : 0 \le x < a, 0 \le y \le b\}$
(C) $R = \{(x, y) : 0 \le x \le a, 0 < y < b\}$ (D) $R = \{(x, y) : 0 < x < a, 0 < y < b\}$

35. In a class of 60 students, 25 students play cricket, 20 students play tennis, and 10 students play both games. Then the number of students who play neither is
(A) 0 (B) 25 (C) 35 (D) 45

36. In a town of 840 persons, 450 persons read Hindi, 300 read English and 200 read both. Then the number of persons who read neither is
(A) 210 (B) 290 (C) 180 (D) 260

37. If $X = \{8^n - 7n - 1 \mid n \in \mathbb{N}\}$ and $Y = \{49n - 49 \mid n \in \mathbb{N}\}$, then
(A) $X \subset Y$ (B) $Y \subset X$ (C) $X = Y$ (D) $X \cap Y = \varnothing$

38. A survey shows that 63% of the people watch a news channel whereas 76% watch another channel. If x% of the people watch both channels, then
(A) $x = 35$ (B) $x = 63$ (C) $39 \le x \le 63$ (D) $x = 39$

39. If the sets A and B are defined as $A = \left\{(x, y) \mid y = \frac{1}{x}, 0 \ne x \in \mathbb{R}\right\}$ and $B = \{(x, y) \mid y = -x, x \in \mathbb{R}\}$, then
(A) $A \cap B = A$ (B) $A \cap B = B$ (C) $A \cap B = \varnothing$ (D) $A \cup B = A$

40. If A and B are two sets, then $A \cap (A \cup B)$ equals
(A) A (B) B (C) $\varnothing$ (D) $A \cap B$

41. If $A = \{1, 3, 5, 7, 9, 11, 13, 15, 17\}$, $B = \{2, 4, ..., 18\}$ and the set $\mathbb{N}$ of natural numbers is the universal set, then $A' \cup ((A \cup B) \cap B')$ is
(A) $\varnothing$ (B) $\mathbb{N}$ (C) A (D) B

42. Let $S = \{x \mid x$ is a positive multiple of 3 less than $100\}$ and $P = \{x \mid x$ is a prime number less than $20\}$. Then $n(S) + n(P)$ is
(A) 34 (B) 31 (C) 33 (D) 30

43. If X and Y are two sets and X' denotes the complement of X, then $X \cap (X \cup Y)'$ is equal to
(A) X (B) Y (C) $\varnothing$ (D) $X \cap Y$

Fill in the blanks

Fill in the blanks in each of Exercises 44 to 51.

44. The set $\{x \in \mathbb{R} : 1 \le x < 2\}$ can be written as _____.

45. When $A = \varnothing$, the number of elements in $P(A)$ is _____.

46. If A and B are finite sets such that $A \subset B$, then $n(A \cup B) =$ _____.

47. If A and B are any two sets, then $A - B$ is equal to _____.

48. The power set of the set $A = \{1, 2\}$ is _____.

49. Given the sets $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{0, 2, 4, 6, 8\}$, the universal set of all three sets A, B and C can be _____.

50. If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 5\}$, $B = \{2, 4, 6, 7\}$ and $C = \{2, 3, 4, 8\}$, then
(i) $(B \cup C)'$ is _____. (ii) $(C - A)'$ is _____.

51. For all sets A and B, $A - (A \cap B)$ is equal to _____.

52. Match the following sets, for all sets A, B and C.

(i)	$((A' \cup B') - A)'$	(a)	$A - B$
(ii)	$[B' \cup (B' - A)]'$	(b)	A

(iii) $(A - B) - (B - C)$
(iv) $(A - B) \cap (C - B)$
(v) $A \times (B \cap C)$
(vi) $A \times (B \cup C)$

(c) B
(d) $(A \times B) \cap (A \times C)$
(e) $(A \times B) \cup (A \times C)$
(f) $(A \cap C) - B$

True or false

State true or false for the statements in each of Exercises 53 to 58.

53. If A is any set, then $A \subset A$.

54. Given that $M = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then $B \not\subset M$.

55. The sets $\{1, 2, 3, 4\}$ and $\{3, 4, 5, 6\}$ are equal.

56. $\mathbb{Q} \cup \mathbb{Z} = \mathbb{Q}$, where $\mathbb{Q}$ is the set of rational numbers and $\mathbb{Z}$ is the set of integers.

57. Let the sets R and T be defined as $R = \{x \in \mathbb{Z} \mid x \text{ is divisible by } 2\}$ and $T = \{x \in \mathbb{Z} \mid x \text{ is divisible by } 6\}$. Then $T \subset R$.

58. Given $A = \{0, 1, 2\}$ and $B = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$, then $A = B$.